

Universal Coset Field Theory: Clock-Field Realizations, Exceptional-Algebra Data, and Controlled Quotient Sectors

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Universal Coset Field Theory (UCFT) is formulated as a bundle-theoretic framework for field theories whose internal kinematics is a homogeneous coset G/H . The data consist of a spacetime base, an H bundle, a coset section, invariant metrics and interaction tensors, matter representations, and an explicit ledger of projection and dynamical assumptions. This separation makes clear what is fixed by representation theory and what must be supplied by dynamics. The benchmark realization is the Hermitian symmetric space $E_6/(\text{Spin}(10) \cdot \text{U}(1))$, whose 32 real tangent directions define clock fields transforming as $\mathbf{16} \oplus \mathbf{\bar{16}}$ under $\text{Spin}(10) \cdot \text{U}(1)$. The framework records the E_6 branching of the $\mathbf{27}$, the induced-gravity heat-kernel mechanism, Green-Schwarz-type axion couplings, heterotic moduli interfaces, and solitonic worldsheet sectors as controlled structures with stated hypotheses. UCFT thereby gives a precise second-iteration language for exceptional coset model building and its low-energy physical interpretations.

I. INTRODUCTION

Exceptional groups and division-algebraic structures give compact descriptions of several recurring facts in high-energy theory: grand-unified embeddings, spinor representations, anomaly-canceling axions, and heterotic string gauge sectors. Those facts are structurally rigid, but they do not by themselves determine a complete physical model. A coset G/H fixes a tangent representation and a class of invariant tensors. A quantum field theory additionally requires a spacetime base, a gauge bundle, matter representations, a vacuum, a potential, a regularization prescription, a symmetry-breaking path, and a rule for matching observables.

Universal Coset Field Theory (UCFT) is introduced as a precise language for this separation. It is not merely a choice of homogeneous space. It is a field-theoretic datum that records which statements follow from coset geometry and which statements require additional hypotheses. This is the central refinement relative to a pure clock-field presentation. The scalar clock fields still play a distinguished role in the E_6 benchmark, but they are now embedded in a more general coset-and-bundle framework that makes the assumptions visible.

The main benchmark is

$$M_{\text{EIII}} = E_6/(\text{Spin}(10) \cdot \text{U}(1)), \quad (1)$$

where the dot denotes the standard central quotient. This compact Hermitian symmetric space has real dimension 32 and complex dimension 16. Its tangent representation is the spinor plus conjugate spinor of $\text{Spin}(10)$, charged under the $\text{U}(1)$ factor. In the clock-field realization, the local coordinates π^α on M_{EIII} are 32 real scalar fields. They act as order parameters for internal symmetry breaking and as coordinates on a curved target space.

The novelty of UCFT is not the isolated observation that E_6 contains $\text{Spin}(10)$, or that $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{\bar{16}}$ under $\text{Spin}(10) \cdot \text{U}(1)$. The novelty is the framework that organizes those facts into an auditable effective field theory. It specifies when a sigma-model term is gauge invariant, when an induced Einstein-Hilbert term follows from heat-kernel coefficients, when a Green-Schwarz-type coupling cancels a factorized anomaly, when a quotient projection can select a chiral sector, and when a solitonic worldsheet sector is actually supported by the topology and dynamics of the vacuum manifold. This improves the explanatory content of the construction: the same algebraic skeleton can support several physical realizations, and UCFT identifies the extra data that distinguishes them.

This paper has four goals. First, it defines UCFT as a bundle-theoretic effective field theory over a specified spacetime. Second, it proves the basic covariance statement: the universal sigma-model, Yang-Mills, Dirac, and Yukawa terms are globally defined when their tensors are H invariant. Third, it develops the $E_6/(\text{Spin}(10) \cdot \text{U}(1))$ clock-field realization, including induced gravity, anomaly couplings, and Standard-Model branchings. Fourth, it incorporates quotient, heterotic-interface, capacity, and solitonic sectors as conditional structures rather than as unqualified consequences of the bare coset.

Throughout the paper, claims are classified as structural, dynamical, or conditional. A structural statement follows from algebra, topology, or representation theory. A dynamical statement follows after a Lagrangian, vacuum, regulator, and field content are fixed. A conditional statement is a theorem only after the required compactification, bundle, projection, or cohomology data are specified. This classification is essential for a publication-level formulation.

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II. UCFT DATA

Let G be a Lie group and $H \subset G$ a closed subgroup. A reductive homogeneous space $M = G/H$ has a vector-space decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}, \quad \text{Ad}(H)\mathfrak{m} \subset \mathfrak{m}. \quad (2)$$

If G/H is symmetric, then $[\mathfrak{m}, \mathfrak{m}] \subset \mathfrak{h}$. A G -invariant metric on M is equivalent to an H -invariant nondegenerate bilinear form on $\mathfrak{m}$.

Definition II.1 (Bare UCFT datum). *A bare UCFT datum is a tuple*

$$\mathcal{D}_0 = (G, H, M = G/H, K, \mathcal{R}, \mathcal{I}), \quad (3)$$

where $K \in (\mathfrak{m}^* \otimes \mathfrak{m}^*)^H$ is an invariant tangent metric, $\mathcal{R}$ is a finite collection of finite-dimensional H representations, and $\mathcal{I}$ is a finite collection of invariant tensors built from $\mathfrak{m}$ and $\mathcal{R}$.

A bare datum is algebraic and kinematic. It is not yet a physical theory. A field theory also needs a base space, gauge bundles, spin data, potentials, and a rule for interpreting sectors.

Definition II.2 (Physical UCFT datum). *A physical UCFT datum over a d -dimensional spacetime X is*

$$\mathcal{D} = (X, g_X, \mathfrak{s}; G, H, K; P_H; \phi, A; R_F, R_S; V, Y; \Pi; \mathcal{L}). \quad (4)$$

Here $(X, g_X, \mathfrak{s})$ is a spacetime manifold with metric and, when fermions are included, spin structure; $P_H \rightarrow X$ is an H principal bundle; $\phi \in \Gamma(P_H \times_H M)$ is the coset field; A is an H connection; R_F and R_S are fermion and scalar representations; V is an invariant potential; Y denotes invariant interaction tensors; Π is a sector-selection or quotient projection when one is specified; and $\mathcal{L}$ is the assumption ledger.

The ledger $\mathcal{L}$ is part of the physical datum. It records every ingredient not fixed by the coset: spacetime dimension and signature, the spin structure, the regulator, nonminimal curvature couplings, the vacuum, the family count, threshold corrections, flavor textures, and any quotient or cohomology calculation. The ledger prevents a representation-theoretic fact from being read as a dynamical prediction.

Definition II.3 (Clock-field realization). *A clock-field realization of a UCFT datum is a physical datum in which the coset section is written locally as*

$$g(x) = \exp[i\pi^\alpha(x)T_\alpha], \quad T_\alpha \in \mathfrak{m}, \quad (5)$$

so that π^α are scalar coordinates on G/H . The fields π^α are called clock fields because they parameterize internal displacement from a chosen symmetric reference configuration.

The term ‘‘clock’’ has no independent physical content unless a dynamics is specified. It labels the scalar order-parameter sector of a nonlinear realization.

III. UNIVERSAL ACTION AND COVARIANCE

The Maurer-Cartan form of a local coset representative is

$$\Omega = g^{-1}dg = \Omega_{\mathfrak{h}} + \Omega_{\mathfrak{m}}. \quad (6)$$

The $\mathfrak{m}$ component defines the vielbein on G/H ; the $\mathfrak{h}$ component defines the composite connection in the ungauged nonlinear realization. When H is gauged by an external connection A , the covariant derivative of the associated coset section is denoted $D_A\phi$.

The universal bosonic sigma-model sector is

$$S_\sigma[\phi, A] = \frac{f^2}{2} \int_X d^d x \sqrt{|g_X|} g_X^{\mu\nu} K_{ab}(\phi) D_\mu \phi^a D_\nu \phi^b - \int_X d^d x \sqrt{|g_X|} V(\phi), \quad (7)$$

where f is the sigma-model normalization. If A is dynamical, the Yang-Mills sector is

$$S_{\text{YM}}[A] = -\frac{1}{4} \sum_i \frac{1}{g_i^2} \int_X d^d x \sqrt{|g_X|} \text{Tr}_i(F_{\mu\nu} F^{\mu\nu}), \quad (8)$$

with the sum over simple and abelian factors. Fermions in R_F have the formal Dirac action

$$S_F = \int_X d^d x \sqrt{|g_X|} i\bar{\psi} \gamma^\mu D_\mu \psi, \quad (9)$$

provided X has the required spin data. Yukawa couplings have the schematic form

$$S_Y = \int_X d^d x \sqrt{|g_X|} (Y_{abc} \psi^a \psi^b \Phi^c + \text{h.c.}), \quad Y \in (R_F^* \otimes R_F^* \otimes R_S^*)^H. \quad (10)$$

Theorem III.1 (Bundle covariance of the UCFT action). *Let $\mathcal{D}$ be a physical UCFT datum. If K , V , the trace forms in Eq. (8), and all interaction tensors Y are H invariant, then*

$$S_{\text{UCFT}} = S_\sigma + S_{\text{YM}} + S_F + S_Y \quad (11)$$

is independent of local trivialization of P_H and invariant under gauge transformations of P_H covering the identity on X .

Proof. On overlaps of local trivializations, fields transform by the relevant representation of an H -valued transition function. The covariant derivative $D_A\phi$ transforms in the associated tangent representation, and contraction with K is invariant because $K \in (\mathfrak{m}^* \otimes \mathfrak{m}^*)^H$. The potential descends to the associated bundle because it is H invariant. Curvatures transform in the adjoint representation, and invariant trace forms make Eq. (8) gauge invariant. The Dirac term is invariant under the induced action on the associated spinor bundle. The Yukawa term patches globally precisely when Y is an H -invariant tensor. Hence the local expressions define a global functional. $\square$

This theorem is a covariance theorem, not a selection theorem. It does not determine X , G , H , the vacuum, the fermion multiplicities, the potential, or numerical low-energy parameters.

IV. EXCEPTIONAL CLOCK-FIELD BENCHMARK

A. Division-algebra and Jordan-algebra input

The exceptional benchmark is based on the real octonions $\mathbb{O}$ and the Albert algebra

$$J_3(\mathbb{O}) = J_3(\mathbb{O}), \quad (12)$$

the Jordan algebra of 3×3 octonion-Hermitian matrices with product

$$X \circ Y = \frac{1}{2}(XY + YX). \quad (13)$$

It has real dimension

$$\dim_{\mathbb{R}} J_3(\mathbb{O}) = 3 + 3 \dim_{\mathbb{R}} \mathbb{O} = 27. \quad (14)$$

The automorphism group preserving the Jordan product is the compact group F_4 . The reduced structure group preserving the cubic norm is of type E_6 [1–5].

Proposition IV.1 (F_4 versus E_6). $F_4 = \text{Aut}(J_3(\mathbb{O}))$ preserves the Jordan product. The compact E_6 used in the benchmark preserves the cubic norm on the **27** and contains $\text{Spin}(10) \cdot \text{U}(1)$ as the isotropy of the Hermitian symmetric space *EIII*.

This distinction is operational. The **27** is the fundamental E_6 representation. The trace-zero 26 is associated with F_4 structure, not with the full E_6 fundamental.

B. EIII and the 32 clock fields

Let

$$M_{\text{EIII}} = E_6/(\text{Spin}(10) \cdot \text{U}(1)). \quad (15)$$

Then

$$\dim_{\mathbb{R}} M_{\text{EIII}} = 78 - (45 + 1) = 32. \quad (16)$$

The adjoint decomposition under $H = \text{Spin}(10) \cdot \text{U}(1)$ is

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}. \quad (17)$$

Hence

$$\mathfrak{m}_{\text{EIII}} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (18)$$

with real dimension 32. The 32 local scalar coordinates π^α in Eq. (5) are the clock fields of the EIII realization.

Theorem IV.2 (EIII tangent representation). *The tangent representation of $E_6/(\text{Spin}(10) \cdot \text{U}(1))$ is Eq. (18). Any further decomposition, for example into spacetime, flavor, axionic, or solitonic sectors, is additional UCFT data and is not implied by EIII alone.*

The fundamental E_6 representation branches as

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}. \quad (19)$$

This is the central representation-theoretic bridge to $\text{Spin}(10)$ grand unification [9–12].

The normalized representation-capacity weights of the **27** are therefore

$$(w_1, w_{10}, w_{16}) = \left(\frac{1}{27}, \frac{10}{27}, \frac{16}{27} \right). \quad (20)$$

These weights are a representation ledger, not a dynamical probability law, unless a capacity principle is added.

C. Local clock-field geometry

For $T_\alpha \in \mathfrak{m}_{\text{EIII}}$, write a local representative as in Eq. (5). The Maurer-Cartan expansion begins

$$\Omega = g^{-1}dg = id\pi^\alpha T_\alpha - \frac{1}{2}\pi^\beta d\pi^\gamma [T_\beta, T_\gamma] + \dots \quad (21)$$

The target metric may be normalized by

$$g_{\alpha\beta}(\pi) = \langle \Omega_\alpha, \Omega_\beta \rangle, \quad g_{\alpha\beta}(0) = \delta_{\alpha\beta}, \quad (22)$$

up to an overall scale convention. The clock-field Lagrangian is

$$\mathcal{L}_\pi = \sqrt{-g_X} \left[\frac{f^2}{2} g_{\alpha\beta}(\pi) D_\mu \pi^\alpha D^\mu \pi^\beta - \xi R \pi^\alpha \pi_\alpha - V(\pi) \right]. \quad (23)$$

Here X is a Lorentzian spacetime chosen as part of the physical datum. The compact target metric $g_{\alpha\beta}$ is not the spacetime metric.

A symmetry-breaking potential can be generated or postulated in the form

$$V(\pi) = \lambda(\pi^\alpha \pi_\alpha - v^2)^2 + \Delta V_{\text{inv}}(\pi), \quad (24)$$

where ΔV_{inv} contains additional H -invariant or discrete-symmetry-controlled operators. A Coleman-Weinberg mechanism may generate such terms when heavy fields couple to the clock sector [16, 17]. This is a dynamical choice, not a consequence of the coset alone.

V. INDUCED GRAVITY IN THE CLOCK-FIELD SECTOR

UCFT can implement Sakharov induced gravity by treating g_X as a background metric at the classical level

and integrating out the clock-field fluctuations [18–23]. Near a local vacuum with $g_{\alpha\beta} \simeq \delta_{\alpha\beta}$, the quadratic fluctuation operator is

$$(\Delta_\pi)_{\alpha\beta} = -\delta_{\alpha\beta} \nabla^2 + \xi R \delta_{\alpha\beta} + \mathcal{M}_{\alpha\beta}^2 + \dots \quad (25)$$

The one-loop effective action is

$$\Gamma_1 = \frac{1}{2} \text{Tr} \log \Delta_\pi = -\frac{1}{2} \int_\epsilon^\infty \frac{ds}{s} \text{Tr} e^{-s\Delta_\pi}, \quad \epsilon \sim \Lambda^{-2}. \quad (26)$$

The heat-kernel expansion in four dimensions is

$$\begin{aligned} \text{Tr} e^{-s\Delta_\pi} &\sim \frac{1}{(4\pi s)^2} \int_X d^4x \sqrt{-g_X} \\ &\times (a_0 + sa_1 + s^2 a_2 + \dots). \end{aligned} \quad (27)$$

For $N_\pi = 32$ real scalar modes, the curvature coefficient gives the induced term

$$\Delta S_{\text{EH}} = \frac{N_\pi f^2 \Lambda^2}{32\pi^2} \left(\frac{1}{6} - \xi \right) \int_X d^4x \sqrt{-g_X} R. \quad (28)$$

up to regulator-dependent finite terms and normalization conventions.

Matching Eq. (28) to $M_{\text{Pl}}^2 \int \sqrt{-g_X} R/2$ yields

$$M_{\text{Pl}}^2 = \frac{N_\pi f^2 \Lambda^2}{16\pi^2} \left(\frac{1}{6} - \xi \right). \quad (29)$$

The sign of the induced Einstein-Hilbert term is positive when $\xi < 1/6$ in this convention.

Proposition V.1 (Radiative stability of the sign condition). *In perturbative scalar field theory in curved space, the beta function for the nonminimal coupling has the form*

$$\mu \frac{d\xi}{d\mu} = \left(\xi - \frac{1}{6} \right) \gamma_\xi + \dots, \quad (30)$$

where γ_ξ is built from dimensionless couplings. If explicit conformal-breaking terms are controlled and $\xi < 1/6$ at the matching scale, the sign of Eq. (29) is stable within the perturbative domain.

Proof. Equation (30) is the standard form of the curved-space renormalization of the scalar curvature coupling. The fixed point at $\xi = 1/6$ is the conformal value for massless scalars. Perturbative corrections proportional to γ_ξ rescale the deviation from this value and cannot flip the sign of $1/6 - \xi$ without driving the theory through $\xi = 1/6$ or out of perturbative control. $\square$

Higher-derivative terms such as R^2 and $R_{\mu\nu} R^{\mu\nu}$ are also induced [24, 25]. They are suppressed at energies $E \ll \Lambda$ by powers of E^2/Λ^2 relative to the leading Einstein-Hilbert term. Thus induced gravity is an effective low-energy sector of a specified UCFT datum. It is not a proof that spacetime or Lorentzian signature follows from M_{EIH} alone.

VI. MATTER, ANOMALY STRUCTURE, AND THE STANDARD-MODEL SKELETON

The standard embedding chain is

$$\begin{aligned} E_6 &\supset \text{Spin}(10) \supset \text{SU}(5) \supset G_{\text{SM}}, \\ G_{\text{SM}} &= \text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y. \end{aligned} \quad (31)$$

up to a finite central quotient. Under $\text{Spin}(10) \supset \text{SU}(5) \times \text{U}(1)_\chi$,

$$\mathbf{16} = \mathbf{10}_{-1} \oplus \bar{\mathbf{5}}_{+3} \oplus \mathbf{1}_{-5}. \quad (32)$$

This is one Standard-Model family plus a right-handed-neutrino singlet in the usual $\text{SO}(10)$ embedding. The additional $\mathbf{10}$ and $\mathbf{1}$ in Eq. (19) are normally interpreted as vectorlike exotics, Higgs-sector candidates, or singlet fields, depending on the physical datum.

The standard tensor products include

$$\mathbf{16} \otimes \mathbf{16} = \mathbf{10}_s \oplus \mathbf{120}_a \oplus \mathbf{126}_s, \quad (33)$$

$$\mathbf{16} \otimes \bar{\mathbf{16}} = \mathbf{1} \oplus \mathbf{45} \oplus \mathbf{210}. \quad (34)$$

Hence $\mathbf{10}$, $\mathbf{120}$, and $\mathbf{126}$ channels are allowed for $\mathbf{1616}$ Yukawa couplings. In a clock-field realization, flavor textures can be encoded by higher-dimensional operators

$$\mathcal{L}_{\text{flav}} \supset \frac{c_{ij}^{(k)}}{\Lambda^k} (\pi^\alpha \mathcal{T}_\alpha)^k \psi_i \psi_j H + \text{h.c.}, \quad (35)$$

where the tensor $\mathcal{T}_\alpha$ and the coefficients $c_{ij}^{(k)}$ must be fixed by the chosen discrete symmetry, vacuum alignment, and representation assignments. This reproduces the logic of Froggatt-Nielsen and discrete-flavor models without claiming a unique texture from E_6 alone [29–31].

A. Green-Schwarz-type axion coupling

A chiral $\text{Spin}(10) \cdot \text{U}(1)$ spectrum can generate factorized gauge and mixed gravitational anomalies. UCFT cancels these anomalies only after the physical datum contains an axionlike clock-field direction with the required shift transformation. Let

$$\theta(\pi) = c_\alpha \pi^\alpha, \quad \delta_\lambda \theta(\pi) = \kappa \lambda. \quad (36)$$

A four-dimensional Green-Schwarz-type term is

$$\begin{aligned} S_{\text{GS}} &= \int_X \theta(\pi) [k_{10} \text{tr}(F_{10} \wedge F_{10}) + k_1 F_1 \wedge F_1 \\ &\quad + k_R \text{tr}(R \wedge R)]. \end{aligned} \quad (37)$$

Proposition VI.1 (Conditional anomaly cancellation). *Assume that the anomaly polynomial of the chiral fermion spectrum factorizes with the same coefficients that appear in Eq. (37), and assume that $\theta(\pi)$ shifts as in Eq. (36). Then the variation of S_{GS} cancels the anomalous variation of the fermion determinant.*

Proof. The anomalous variation of the fermion determinant is determined by the descent of the factorized anomaly polynomial. The shift of θ multiplies the same characteristic classes. Equality of the coefficients cancels the local anomaly term by term. $\square$

This statement is the correct field-theoretic version of the axion mechanism. It requires a specified chiral spectrum and a specified shifting scalar. It does not follow from the bare EIII coset without those data.

B. High-scale weak-angle normalization

With canonical SU(5) or Spin(10) hypercharge normalization and a unified gauge coupling at a scale M_U ,

$$\alpha_1 = \frac{5}{3}\alpha_Y, \quad g_1 = g_2 = g_3 = g_U, \quad (38)$$

and therefore

$$\sin^2 \theta_W(M_U) = \frac{g_Y^2}{g_2^2 + g_Y^2} = \frac{3}{8}. \quad (39)$$

This is a high-scale normalization condition. The low-energy value depends on the renormalization-group flow, threshold corrections, and field content [26–28].

VII. HETEROTIC INTERFACE AND QUOTIENT DATA

The original clock-field motivation is sharpened in UCFT as an interface between a four-dimensional coset field theory and heterotic compactification data. The heterotic string contains a ten-dimensional Green-Schwarz structure

$$H_3 = dB_2 + \omega_{\text{YM}} - \omega_L, \quad dH_3 = \text{tr}(R \wedge R) - \text{tr}(F \wedge F), \quad (40)$$

which reduces to axionic anomaly-canceling couplings in four dimensions [32–36, 46, 47].

A compactification interface is not part of the bare EIII coset. It is additional UCFT data

$$\mathcal{C} = (\hat{Q}, V, \mathcal{W}, H_3, \mathcal{F}, \sigma), \quad (41)$$

where $\hat{Q}$ is a compact space or quotient, V is a holomorphic or smooth gauge bundle, $\mathcal{W}$ denotes Wilson-line data, H_3 denotes flux data, $\mathcal{F}$ denotes stabilization data, and σ denotes a possible involution or projection.

The scalar moduli count in a heterotic model before stabilization has the schematic form

$$N_{\text{moduli}} = 2h^{2,1}(\hat{Q}) + h^{1,1}(\hat{Q}) + 2 \dim H^1(\hat{Q}, V) + 2, \quad (42)$$

with the last term representing the axion-dilaton sector in the simplest setting. Fluxes, nonperturbative effects, and discrete identifications can lift or identify directions

[37–45]. A UCFT-heterotic interface exists locally when the residual moduli manifold $\mathcal{M}_{\text{het}}$ has real dimension 32 and admits a local map

$$\pi^\alpha = \pi^\alpha(\Phi^I), \quad I = 1, \dots, 32, \quad (43)$$

with metric pullback

$$G_{IJ}(\Phi) = f^2 \frac{\partial \pi^\alpha}{\partial \Phi^I} g_{\alpha\beta}(\pi) \frac{\partial \pi^\beta}{\partial \Phi^J}. \quad (44)$$

Proposition VII.1 (Local heterotic interface). *If Eqs. (43) and (44) hold on a patch of the stabilized moduli space, and if the shifting axion direction matches Eq. (37), then the UCFT clock sector and the residual heterotic moduli sector have the same local scalar kinetic geometry and the same local anomaly-canceling axion structure.*

This is the technically precise content of the moduli-matching claim. It is stronger than a numerical equality of dimensions but weaker than a global duality. Global equivalence requires transition functions, discrete identifications, flux quantization, and anomaly data to agree on overlaps.

Example VII.2 (A quotient Calabi-Yau interface). *A common test case is a freely acting $\mathbb{Z}_5$ quotient of a quintic Calabi-Yau, together with a bundle and Wilson line that break an E_8 factor through an E_6 stage to a Spin(10) · U(1) sector. In UCFT this example should be presented as a conditional interface: after the quotient, bundle cohomology, flux constraints, and nonperturbative stabilization are specified and computed, one checks whether exactly 32 real moduli survive and whether Eq. (44) holds. The number 32 is then a result of the chosen compactification data, not of EIII by itself.*

VIII. SOLITONIC AND WORLDSHEET SECTORS

Solitonic sectors are useful, but they must be stated with correct topology. For a simply connected compact group E_6 and connected isotropy Spin(10) · U(1), the compact Hermitian symmetric space $E_6/(\text{Spin}(10) \cdot \text{U}(1))$ is simply connected. The long exact homotopy sequence instead gives a nontrivial two-cycle associated with the U(1) factor. Thus a bare EIII sigma model does not by itself guarantee stable cosmic strings through $\pi_1(M_{\text{EIII}})$.

Stringlike solitons can still appear in a physical UCFT datum through additional structure: a gauged Abelian Higgs sector, a quotient with nontrivial fundamental group, a vacuum manifold distinct from the target, an embedded vortex supported by a potential, or a compactification sector with nontrivial one-cycles.

Definition VIII.1 (Vortex sector). *A UCFT vortex sector is a tuple*

$$\mathcal{V} = (\mathcal{N}, \varphi, A_v, V_v, \iota), \quad (45)$$

where $\mathcal{N}$ is the vacuum manifold of the relevant low-energy fields, φ is the winding scalar or multiplet, A_v is any gauge field required for finite tension, V_v is the vortex-supporting potential, and $\iota : \mathcal{N} \rightarrow M$ embeds the vortex sector into the UCFT target data.

For a straight static vortex along the z direction, a typical ansatz is

$$\varphi(r, \theta) = v f(r) e^{in\theta}, \quad A_\theta(r) = \frac{n}{q} a(r), \quad (46)$$

with $f(0) = a(0) = 0$ and $f(\infty) = a(\infty) = 1$. Its tension is finite when the gauge and scalar profiles satisfy the usual Nielsen-Olesen boundary conditions [50–52].

Localized fluctuations around a vortex can be expanded as

$$\delta\pi^\alpha(t, z, r, \theta) = \sum_n \psi_n^\alpha(t, z) \chi_n(r, \theta), \quad (47)$$

where the transverse modes satisfy a self-adjoint eigenvalue problem

$$\mathcal{O}\chi_n = \mu_n^2 \chi_n. \quad (48)$$

Normalizable zero modes define a two-dimensional worldsheet theory. If localized fermion zero modes are present, their net chirality is controlled by an index theorem or Callan-Harvey inflow mechanism [53, 54].

Proposition VIII.2 (Conditional worldsheet current algebra). *Assume that a UCFT vortex sector exists, that its normalizable zero modes transform under $\text{Spin}(10) \cdot \text{U}(1)$, and that the operator products of the corresponding currents close without anomaly. Then the worldsheet currents satisfy an affine algebra*

$$J^A(z) J^B(w) \sim \frac{k\delta^{AB}}{(z-w)^2} + \frac{if^{ABC} J^C(w)}{z-w}. \quad (49)$$

For level $k = 1$, the associated Sugawara stress tensor has central charge

$$c = \frac{k \dim G}{k + g^\vee} \quad (50)$$

for each simple factor, plus the abelian contribution.

For $\text{SO}(10)$ at level one, $g^\vee = 8$ and $\dim \text{SO}(10) = 45$, so $c_{\text{SO}(10)_1} = 45/9 = 5$. Additional free fermions, bosons, lattice sectors, or hidden gauge sectors are required to reproduce the full heterotic central-charge bookkeeping [55–58]. This is another place where the ledger is essential: affine symmetry is a theorem only after the zero-mode spectrum and operator algebra are derived.

IX. CAPACITY, MIRROR OPERATIONS, AND PROJECTION SECTORS

The first iteration of clock-field language used mirror, capacity, and sector-projection terminology to organize algebraic sectors. UCFT retains this terminology but assigns it a precise status.

Definition IX.1 (Capacity ledger). *Let $R = \oplus_i R_i$ be an H -module decomposition into isotypic blocks. The representation capacity ledger is*

$$w_i = \frac{\dim R_i}{\dim R}, \quad \sum_i w_i = 1. \quad (51)$$

A dynamical capacity principle is an additional variational postulate whose stationary point is required to reproduce or replace these weights.

A useful finite-dimensional dual functional is

$$C_m(\omega) = - \sum_{a \in \mathcal{A}} \mu(a) \log \left[\sum_i \exp(c_i(a) - \omega_i) \right] - \sum_i m_i \omega_i, \quad \sum_i \omega_i = 0. \quad (52)$$

Its Hessian is the negative of a weighted softmax covariance matrix. Therefore C_m is concave on the affine slice and strictly concave when the covariance is nondegenerate there. At an interior stationary point,

$$\sum_{a \in \mathcal{A}} \mu(a) p_i(a; \omega) = m_i, \quad p_i(a; \omega) = \frac{\exp(c_i(a) - \omega_i)}{\sum_j \exp(c_j(a) - \omega_j)}. \quad (53)$$

Mirror operations are similarly structural until dynamical content is assigned. Examples include complex conjugation of representations, octonionic conjugation, exchange of two product factors, and the EIII pairing $\mathbf{16}_{-3} \leftrightarrow \mathbf{16}_{+3}$.

Definition IX.2 (Sector projection). *Let σ be an involution acting on a vector bundle $E \rightarrow X$ and commuting with the relevant H action. The sector projectors are*

$$\Pi_\pm = \frac{1}{2}(\text{id} \pm \sigma), \quad \Pi_\pm^2 = \Pi_\pm, \quad \Pi_+ \Pi_- = 0. \quad (54)$$

Proposition IX.3 (Gauge-compatible projection). *If σ commutes with the H action and with the kinetic operators, then $\Pi_\pm E$ are gauge-compatible subbundles. The projected action remains gauge invariant when interaction tensors are restricted to invariant tensors on the projected sector.*

Proof. The projector identities follow from $\sigma^2 = \text{id}$. Commutation with H implies stability under gauge transformations. Commutation with the kinetic operators prevents kinetic mixing between the projected sectors. The bundle-covariance theorem then applies after restriction. $\square$

Corollary IX.4 (Conditional family-slot statement). *Let $\hat{Q}$ be a compact complex manifold, $V_{16} \rightarrow \hat{Q}$ a holomorphic bundle transforming as a $\mathbf{16}$ of $\text{Spin}(10)$, and σ an involution lifting to the Dolbeault complex. If*

$$\dim H^1(\hat{Q}, V_{16})^\eta = 3, \quad \eta \in \{+1, -1\}, \quad (55)$$

then the selected projected sector contains three $\mathbf{16}$ family slots. The number three is a theorem of the specified compactification, bundle, and projection data; it is not a theorem of the bare EIII coset.

TABLE I. Standard correspondence between $H_2(A)$ and Minkowski spaces. The ordinary four-dimensional case is $H_2(\mathbb{C})$, not $H_2(\mathbb{H})$.

A	$\dim_{\mathbb{R}} A$	$\dim_{\mathbb{R}} H_2(A)$	model
$\mathbb{R}$	1	3	$\mathbb{R}^{1,2}$
$\mathbb{C}$	2	4	$\mathbb{R}^{1,3}$
$\mathbb{H}$	4	6	$\mathbb{R}^{1,5}$
$\mathbb{O}$	8	10	$\mathbb{R}^{1,9}$

X. DIVISION ALGEBRAS AND THE SPACETIME CAVEAT

The division-algebra ladder is relevant but frequently misread. Hurwitz’s theorem states that the finite-dimensional real normed division algebras with identity are

$$\mathbb{R}, \quad \mathbb{C}, \quad \mathbb{H}, \quad \mathbb{O}, \quad (56)$$

of dimensions 1, 2, 4, 8 respectively [1–3]. The Cayley-Dickson sequence continues formally to the sedenions, but zero divisors appear and the normed-division property is lost.

For $A \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$, the Hermitian 2×2 matrices

$$X = \begin{pmatrix} t+x & y \\ y^* & t-x \end{pmatrix}, \quad t, x \in \mathbb{R}, \quad y \in A, \quad (57)$$

have determinant

$$\det X = t^2 - x^2 - |y|^2, \quad (58)$$

with Lorentzian signature $(1, \dim_{\mathbb{R}} A + 1)$.

The quaternionic block $H_2(\mathbb{H})$ is six-dimensional with signature $(1, 5)$. Four-dimensional Minkowski space is the complex block $H_2(\mathbb{C})$, with $\text{Spin}(1, 3) \simeq \text{SL}(2, \mathbb{C})$. A UCFT may choose a four-dimensional base X , but that choice is not derived from the compact internal EIII metric.

XI. DISCUSSION

UCFT provides a technical way to keep the strongest features of the exceptional-algebra program while removing ambiguous inference. The E_6 benchmark remains powerful: it supplies the 32 clock-field directions, the **27** decomposition, the $\text{Spin}(10)$ family embedding, the canonical unification normalization, and natural interfaces with heterotic compactification data. The framework deepens the interpretation by explaining how these structures fit together: the coset supplies kinematics, the bundle supplies gauge covariance, the Lagrangian supplies dynamics, the heat-kernel calculation supplies induced gravity, the anomaly polynomial supplies the axion coupling, and quotient or compactification data supply family selection.

This organization also identifies the limits of the construction. The bare coset does not determine a Lorentzian spacetime, a unique vacuum, a cosmic-string network, three families, the measured electroweak angle, or numerical masses. Those can be studied inside UCFT, but only after the relevant physical datum is specified.

The resulting second-iteration picture is therefore more precise and more useful than a catalogue of coincidences. It states a general mathematical framework and then shows how the clock-field realization of $E_6/(\text{Spin}(10) \cdot \text{U}(1))$ fits into that framework. This is the level at which UCFT can function as a professional model-building language: every claimed prediction is tied to an explicit entry in the datum, and every structural theorem states its hypotheses.

The next technical steps are concrete. One should classify invariant potentials and Yukawa tensors for the EIII datum, compute anomaly polynomials for proposed chiral spectra, construct explicit quotient or compactification interfaces with verified cohomology, derive any solitonic worldsheet operator algebra from the four-dimensional fluctuation problem, and formulate numerical phenomenology only after an input count and threshold scheme are fixed.

XII. CONCLUSION

Universal Coset Field Theory is a field-theoretic framework for homogeneous internal spaces and their physical realizations. Its central datum combines a coset G/H , an H bundle, invariant tensors, matter representations, potentials, projections, and an explicit assumption ledger. For the EIII benchmark $E_6/(\text{Spin}(10) \cdot \text{U}(1))$, the framework gives a 32-real-dimensional clock-field sector, the tangent representation $\mathbf{16} \oplus \overline{\mathbf{16}}$, the E_6 fundamental branching $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$, and a controlled path to $\text{Spin}(10)$ and Standard-Model embeddings.

The same formalism accommodates induced gravity, Green-Schwarz-type anomaly cancellation, heterotic moduli interfaces, capacity ledgers, quotient projections, and solitonic worldsheet sectors. Each is stated with the hypotheses required for it to be a theorem. This yields a direct and publication-level version of UCFT: a novel framework that uses exceptional coset geometry to organize unification models while separating structural consequences from dynamical assumptions.

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DATA AVAILABILITY

No data were created or analyzed in this theoretical study.

Appendix A: Assumption ledger template

A UCFT manuscript should include a ledger separating at least three classes of entries. Structural entries are fixed by the coset and representation theory: G/H , $\mathfrak{m}$, invariant tensors, group branchings, and topological facts. Dynamical entries require a Lagrangian and regulator: the potential, vacuum, induced gravitational coupling, RG flow, threshold corrections, and mass matrices. Conditional entries require extra geometric or sector data: quotient projections, compactification cohomology, flux stabilization, solitonic sectors, and worldsheet CFTs.

For the EIII benchmark, the following facts are structural:

$$\dim_{\mathbb{R}} E_6/(\text{Spin}(10) \cdot \text{U}(1)) = 32, \quad (\text{A1})$$

$$\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (\text{A2})$$

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}. \quad (\text{A3})$$

The following are not structural consequences of the bare coset: the choice of four-dimensional Lorentzian space-time, three chiral families, a cosmic-string network, inflation, dark matter, proton decay rates, CKM or PMNS matrices, and measured low-energy couplings.

Appendix B: Reference normalization for $\sin^2 \theta_W$

In canonical $\text{SU}(5)$ normalization,

$$\alpha_1 = \frac{5}{3}\alpha_Y, \quad g_1 = g_2 = g_3 = g_U \quad (\text{B1})$$

at the unification scale. Therefore $g_Y^2 = (3/5)g_U^2$ and $g_2^2 = g_U^2$, so

$$\sin^2 \theta_W(M_U) = \frac{g_Y^2}{g_2^2 + g_Y^2} = \frac{3/5}{1 + 3/5} = \frac{3}{8}. \quad (\text{B2})$$

This is independent of low-energy threshold corrections and should be presented as a high-scale normalization condition.

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